

HANAMANTHAGOUDA POLICEPATIL

+91 7829580864 | 23110126@iitgn.ac.in | [SSRN](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

Indian Institute of Technology Gandhinagar (IITGN)
Dual B.Tech. in Computer Science and Materials Science & Engineering

Jul '23 – May '28
CPI: 8.0 / 10.0

RESEARCH EXPERIENCE

Research Intern | Indian Institute of Science (IISc), Bengaluru

May '25 – Jul '25 (Onsite)

Advisor: Prof. Abhishek Kumar Singh | Materials Research Centre (MRC)

Bengaluru, India

- **Multimodal Foundation Models:** Designed an architectural pipeline for domain-adaptive foundation models (CAST, CLIP, DAPT) unifying unstructured scientific literature with structural crystal lattice data.
- **Instruction Fine-Tuning:** Fine-tuned LLaMA-3 8B using QLoRA and Unsloth on specialized instruction datasets for property screening; optimized context lengths and hyperparameters, increasing classification accuracy from a 52.9% baseline to **90.2%** (metal vs. non-metal). [\[GitHub\]](#)
- **Domain-Adaptive Pre-Training (DAPT):** Executed target-domain continual pre-training (CPT) on an extensive scientific corpus using Unsloth; decreased test perplexity to **9.48** and integrated models onto Hugging Face for automated scientific inference. [\[GitHub\]](#)
- **Evaluation Pipelines:** Built reproducible automated evaluation benches and tokenization workflows for downstream zero-shot and few-shot materials property prediction.

PUBLICATIONS

Real-Time Win Probability Estimation in Twenty20 Cricket: A State-Space Approach

2025

Co-authors: Prof. Prateek Sharma (USC) & Prof. Amit Arora (IITGN) | Submitted to *Journal of Quantitative Analysis in Sports* | [\[SSRN\]](#)

- **State-Space Framework:** Formulated a ball-by-ball continuous-discrete state-space framework tracking three latent match dynamics: Batting Strength (S_t), Scoring Momentum (M_t), and Match Pressure (P_t), updated recursively via **Kalman filtering**.
- **Empirical Calibration:** Evaluated across **1,161 Indian Premier League (IPL) matches** spanning 18 seasons (2008–2025); benchmarked against logistic regression, random forests, and gradient boosting, delivering competitive predictive accuracy with strict temporal coherence.
- **Latent Trajectories:** Proved momentum as the earliest indicator of chase success, pressure as the most persistent discriminator between outcomes, and batting strength as a late-innings confirmation signal.

The Crowding Toll on Rented Cognition

2025

Co-authors: Prof. Prateek Sharma (USC) | Submitted to the *American Finance Association (AFA) Annual Meeting* | [\[Link\]](#)

- **Rented Cognition & Common Shocks:** Examined the delegation of cognitive tasks (screening, pricing, forecasting) to foundation models; proved that while human cognitive errors wash away, rented cognition carries a vendor-level common error component, turning procurement into a common macroeconomic risk factor.
- **Endogenous Factor Structure:** Derived the equilibrium "crowding toll" showing that exposure prices rise with vendor clientele; adopting firms become safer and riskier at once (falling idiosyncratic volatility but surging systematic volatility and higher cost of capital).
- **Concentration-Driven Risk:** Factorized aggregate variance as the product of squared total delegation and vendor concentration ($\text{Risk} \propto (\delta_{\text{tot}})^2 \times \text{HHI}$); demonstrated that capability gains raise systematic risk (**+32%** in calibration) primarily by relocating delegation to frontier vendors rather than increasing total adoption.

Adopt, Wait, or Learn? Computational Capital and AI Investment

2025

Co-authors: Prof. Prateek Sharma (USC) | Submitted to India Finance Conference (IFC'26, IIM Bangalore) | [\[SSRN\]](#)

- **Dynamic Investment Framework:** Formulated a dynamic model combining real options, internal capital markets, and organizational learning to analyze corporate adoption timing and token-budget allocation across competing business units.
- **Option Value of Waiting:** Demonstrated that computational token consumption generates organizational capital (reusable prompts, automated agents, workflows), creating a key trade-off between delaying adoption and postponing knowledge accumulation.
- **Calibration & Mechanism Design:** Calibrated using enterprise GitHub production data; proved that decentralized allocation mechanisms strictly outperform centralized capital rationing under reporting noise and compute scarcity.

PROJECTS

Scientific Machine Learning for Physical Systems (Physical PDEs)

2024

Independent Deep Learning Researcher

PyTorch

- **Fourier Neural Operators (FNOs) for Turbulence:** Implemented an FNO architecture in PyTorch to simulate 2D Navier-Stokes turbulence. Integrated a targeted noise-injection strategy during autoregressive rollouts, stabilizing long-term trajectories and reducing validation error to **1.04%**. [\[GitHub\]](#)
- **Physics-Informed Neural Networks (PINNs):** Formulated PINNs in PyTorch to solve forward field predictions and inverse parameter identification for the 1D Burgers' equation; estimated latent PDE parameters (λ_1, λ_2) from noisy measurements with under **0.22%** error. [\[GitHub\]](#)

SKILLS

Programming Languages: Python, C++, C, SQL, Bash

Machine Learning & Libraries: PyTorch, Hugging Face, Scikit-Learn, Unsloth, NumPy, Pandas, SciPy, Matplotlib

Developer Tools & Platforms: Git, Linux/Unix, Docker, Jupyter

LEADERSHIP

Class Representative (MSE '27) | IIT Gandhinagar

2024-25

Primary academic liaison representing 40+ undergraduate students before departmental faculty; coordinated academic schedules, examinations, and feedback channels.